

Riqiang Gao, Ph.D.

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Research Interest

2015–now 📌 I am interested in artificial intelligence, especially its applications in healthcare. My dream is to develop “gentle-and-strict” models that are 1) easy to implement and user-friendly (*gentle*), and 2) motivated by practical challenges and theoretically solid (*strict*).¹

Research Experiences

10/2025 – now 📌 Senior Staff AI Scientist, *Deep (Reinforcement) Learning for Healthcare, Radiotherapy Planning*, Siemens Healthineers

01/2025 – 10/2025 📌 Staff AI Scientist, *Deep (Reinforcement) Learning for Healthcare, Radiotherapy Planning*, Siemens Healthineers

04/2022 – 12/2024 📌 Senior AI Scientist, *Deep (Reinforcement) Learning for Healthcare, Radiotherapy Planning*, Siemens Healthineers

08/2018 – 03/2022 📌 Research Assistant, *Lung Cancer Risk Estimation*, Vanderbilt University, Advisor: Prof. Bennett Landman

05/2021 – 09/2021 📌 Research Intern, *Anomaly Detection with Generative Models*, Siemens Healthineers, Mentor: Dr. Zhoubing Xu

09/2015 – 05/2018 📌 Research Assistant, *Face Recognition and Computer Vision*, Tsinghua University

03/2018 – 06/2018 📌 Research Intern, *Whole Slide Image Analysis*, Insight Technology

06/2017 – 09/2017 📌 Research Intern, *Clothes Detection*, YouTu X-Lab of Tencent

Industrial Management Experiences

10/2025 – now 📌 Project Manager, *AI Based Planning for Brachytherapy*, Siemens Healthineers

10/2023 – now 📌 Project Manager, *Automated AI Planning - RapidPlan 3D*, Siemens Healthineers

10/2023 – 10/2025 📌 Project Manager, *Automated AI Planning - RL optimization*, Siemens Healthineers

08/2023 – 11/2023 📌 Project Manager, *E2E Auto AI Planning with Front End*, Siemens Healthineers

04/2023 – 09/2023 📌 Project Manager, *Automated AI Planning- Accelerated Dose and Leaf Sequencing Modules*, Siemens Healthineers

Education

Degree	Major	University	Period	GPA rank	GPA+award
Ph.D.	Computer Sci.	Vanderbilt Univ.	08/2018 - 03/2022	N/A	N/A
M.E.	Electronics Eng.	Tsinghua Univ.	09/2015 - 07/2018	2 / 54	N/A
B.E.	Communi. Eng.	Central South Univ.	09/2011 - 07/2015	2 / 163	1 / 163

Selected Honors & Awards

2025 📌 NeurIPS 2025 GenAI4Health Best Paper Award (first author)

¹this resume is updated at 05/2026

Selected Honors & Awards (continued)

	1st Winner Team of PANORAMA Challenge
2024	Innovation Excellence Award at Siemens Healthineers (first-author)
2021	MICCAI Traveling Award, now STAR Award (first-author)
	C.F. Chen Best Paper Award (with 5,000 USD), Vanderbilt University (first-author)
	RFW Best Student Paper Award Finalist, SPIE-MI 2021 (first-author) (< 2%)
2020	RFW Best Student Paper Award Finalist, SPIE-MI 2020 (first-author) (< 2%)
	Honorable Mention Poster Award, SPIE-MI 2020 (Mentor & Presenter) (< 5%)
2015	Member of Outstanding Deeds Report (10 selected across all majors, < 0.5%)
	Outstanding Graduate in Hunan Province, China (2%)
2014	Meritorious in Mathematical Contest in Modeling of USA (team-leader) (10%)
	Pacemaker to Merit Student of CSU (30 selected across all majors, < 0.5%)
2013	First Prize in National Mathematics Competitions (not-math-major) of China (Rank 17 in China, Rank 1 in Hunan Province, < 0.02%)
2012 – 2022	10+ Scholarships including Outstanding Scholarship (CSU, < 1%), National Scholarship (CSU, < 2%), First Prize Scholarship (THU, 10%), Dean's Graduate Fellowship (VU).

Selected Publications (Contact Author *)

- 1 Arberet, S., **Gao, Riqiang**, Kraus, M., Ghesu, F. C., Verbakel, W., Diallo, M., Magliari, A., Karuppusamy, V., Beriwal, S., Kamen, A., et al. (2026). Ai end-to-end radiation treatment planning under one second. *arXiv preprint arXiv:2603.06338*.
- 2 Liu, H., Georgescu, B., Zhang, Y., Yoo, Y., Baumgartner, M., **Gao, Riqiang**, Wang, J., Gibson, E., Comaniciu, D., & Grbic, S. (2026). Revisiting 2d foundation models for scalable 3d medical image classification. *IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- 3 Poles, I., Arberet, S., **Gao, Riqiang**, Kraus, M., Santambrogio, M. D., Ghesu, F. C., Kamen, A., & Comaniciu, D. (2026). Learning to optimize radiotherapy plans via fluence maps diffusion model generation and lstm-based optimization. *International Conference on Medical Image Computing and Computer-Assisted Intervention*.
- 4 Wang, Y., Li, Z., Liu, H., Arberet, S., Kraus, M., Zhou, Y., Ghesu, F.-C., Comaniciu, D., Kamen, A., & **Gao, Riqiang***. (2026). Any2any 3d diffusion models with knowledge transfer: A radiotherapy planning study. *IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- 5 Arberet, S., Ghesu, F. C., **Gao, Riqiang**, Kraus, M., Sackett, J., Kuusela, E., & Kamen, A. (2025). A beam eye view to fluence maps 3d network for ultra fast vmat radiotherapy planning. *Medical Physics*.
- 6 Liu, H., **Gao, Riqiang**, & Grbic, S. (2025). Ai-assisted early detection of pancreatic ductal adenocarcinoma on contrast-enhanced ct. *arXiv preprint arXiv:2503.10068*.
- 7 **Gao, Riqiang**, Arberet, S., Kraus, M., Comaniciu, D., Liu, H., Verbakel, W., Ghesu, F. C., & Kamen, A. (2025). Demo: Generative ai helps radiotherapy planning with user preference. *NeurIPS GenAI4Health*.
- 8 **Gao, Riqiang**, Diallo, M., Liu, H., Magliari, A., Sackett, J., Verbakel, W., Meyers, S., Zarepisheh, M., Mcbeth, R., Arberet, S., Kraus, M., Ghesu, F. C., & Kamen, A. (2025). Automating high quality rt planning at scale. *arXiv preprint arXiv:2501.11803*.
- 9 Liu, H., Xu, Z., **Gao, Riqiang**, Li, H., Wang, J., Chabin, G., Oguz, I., & Grbic, S. (2024). Cosst: Multi-organ segmentation with partially labeled datasets using comprehensive supervisions and self-training. *IEEE Transactions on Medical Imaging*.

- 10 **Gao, Riqiang**, Ghesu, F.-C., Arberet, S., Basiri, S., Kuusela, E., Kraus, M., Comaniciu, D., & Kamen, A. (2024). Multi-agent reinforcement learning meets leaf sequencing in radiotherapy. *International Conference on Machine Learning*.
- 11 Li, T. Z., Hin Lee, H., Xu, K., **Gao, Riqiang**, Dawant, B. M., Maldonado, F., Sandler, K. L., & Landman, B. A. (2023). Quantifying emphysema in lung screening computed tomography with robust automated lobe segmentation. *Journal of Medical Imaging*.
- 12 Li, T. Z., Still, J. M., Xu, K., Lee, H. H., Cai, L. Y., Krishnan, A. R., **Gao, Riqiang**, Khan, M. S., Antic, S., Kammer, M., et al. (2023). Longitudinal multimodal transformer integrating imaging and latent clinical signatures from routine ehers for pulmonary nodule classification. *International Conference on Medical Image Computing and Computer-Assisted Intervention*.
- 13 **Gao, Riqiang**, Lou, B., Xu, Z., Comaniciu, D., & Kamen, A. (2023). Flexible- c^m gan: Towards precise 3d dose prediction in radiotherapy. *IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- 14 Xu, K., Khan, S. M., Li, T., **Gao, Riqiang** et al. (2023). Ai body composition in lung cancer screening: Added value beyond lung cancer detection. *Radiology*.
- 15 Xu, K., Li, T., Khan, M. S., **Gao, Riqiang**, Antic, S. L., Huo, Y., Sandler, K. L., Maldonado, F., & Landman, B. A. (2023). Body composition assessment with limited field-of-view computed tomography: A semantic image extension perspective. *Medical Image Analysis*.
- 16 Yu, X., Yang, Q., Zhou, Y., Cai, L. Y., **Gao, Riqiang**, Lee, H. H., Li, T., Bao, S., Xu, Z., Lasko, T. A., et al. (2023). Unest: Local spatial representation learning with hierarchical transformer for efficient medical segmentation. *Medical Image Analysis*.
- 17 **Gao, Riqiang**, Li, T., Tang, Y., Xu, K., Khan, M., Kammer, M., Antic, S. L., Deppen, S., Huo, Y., Lasko, T. A., et al. (2022). Reducing uncertainty in cancer risk estimation for patients with indeterminate pulmonary nodules using an integrated deep learning model. *Computers in Biology and Medicine*.
- 18 **Gao, Riqiang**, Xu, Z., Chabin, G., Mansoor, A., Ghesu, F.-C., Georgescu, B., Landman, B. A., & Grbic, S. (2022). You may need both good-gan and bad-gan for anomaly detection. *Technique Report (not peer-reviewed publication)*.
- 19 Tang, Y., **Gao, Riqiang**, Han, S., Chen, Y., Gao, D., Nath, V., Bermudez, C., Savona, M. R., Bao, S., Lyu, I., et al. (2021). Body part regression with self-supervision. *IEEE Transactions on Medical Imaging*.
- 20 Tang, Y., **Gao, Riqiang**, Lee, H., Yang, Q., Yu, X., Zhou, Y., Bao, S., Huo, Y., Spraggins, J., Virostko, J., et al. (2021). Pancreas ct segmentation by predictive phenotyping. *International Conference on Medical Image Computing and Computer-Assisted Intervention*.
- 21 **Gao, Riqiang**, Tang, Y., Khan, M. S., Xu, K., Paulson, A. B., Sullivan, S., Huo, Y., Deppen, S., Massion, P. P., Sandler, K. L., & Landman, B. A. (2021). Cancer risk estimation combining lung screening ct with clinical data elements. *Radiology: Artificial Intelligence*.
- 22 **Gao, Riqiang**, Tang, Y., Xu, K., Kammer, M. N., Antic, S. L., Deppen, S., Sandler, K. L., Massion, P. P., Huo, Y., & Landman, B. A. (2021). Deep multi-path network integrating incomplete biomarker and chest ct data for evaluating lung cancer risk. *Medical Imaging: Image Processing*.
- 23 **Gao, Riqiang**, Tang, Y., Xu, K., Lee, H. H., Deppen, S., Sandler, K., Massion, P., Lasko, T. A., Huo, Y., & Landman, B. A. (2021). Lung cancer risk estimation with incomplete data: A joint missing imputation perspective. *International Conference on Medical Image Computing and Computer-Assisted Intervention*.
- 24 Tang, Y., **Gao, Riqiang**, Han, S., Chen, Y., Gao, D., Nath, V., Bermudez, C., Savona, M. R., Abramson, R. G., Bao, I., Shunxing Lyu, Huo, Y., & Landman, B. A. (2020a). High-resolution 3d abdominal segmentation with random patch network fusion. *Medical Image Analysis*.

- 25 **Gao, Riqiang**, Huo, Y., Bao, S., Tang, Y., Antic, S. L., Epstein, E. S., Deppen, S., Paulson, A. B., Sandler, K. L., Massion, P. P., & Landman, B. A. (2020). Multi-path xd recurrent neural networks for collaborative image classification. *Neurocomputing*.
- 26 **Gao, Riqiang**, Li, L., Tang, Y., Antic, S. L., Paulson, A. B., Huo, Y., Sandler, K. L., Massion, P. P., & Landman, B. A. (2020). Deep multi-task prediction of lung cancer and cancer-free progression from censored heterogenous clinical imaging. *Medical Imaging: Image Processing*, 11313, 113130D.
- 27 **Gao, Riqiang**, Tang, Y., Xu, K., Huo, Y., Bao, S., Antic, S. L., Epstein, E. S., Deppen, S., Paulson, A. B., Sandler, K. L., Massion, P. P., & Landman, B. A. (2020). Time-distanced gates in long short-term memory networks. *Medical Image Analysis (C.F. Chen best paper (VU))*.
- 28 Yang, Y., **Gao, Riqiang ***, Tang, Y., Antic, S. L., Deppen, S., Huo, Y., Sandler, K. L., Massion, P. P., & Landman, B. A. (2020). Internal-transfer weighting of multi-task learning for lung cancer detection. *Medical Imaging: Image Processing*, 11313, 1131323.
- 29 **Gao, Riqiang**, Huo, Y., Bao, S., Tang, Y., Antic, S. L., Epstein, E. S., Balar, A. B., Deppen, S., Paulson, A. B., Sandler, K. L., et al. (2019). Distanced lstm: Time-distanced gates in long short-term memory models for lung cancer detection, 310–318.
- 30 **Gao, Riqiang**, Yang, F., Yang, W., & Liao, Q. (2018). Margin loss: Making faces more separable. *IEEE Signal Processing Letters*, 25(2), 308–312.
- 31 Yang, F., Yang, W., **Gao, Riqiang**, & Liao, Q. (2017). Discriminative multidimensional scaling for low-resolution face recognition. *IEEE Signal Processing Letters*, 25(3), 388–392.
- 32 Yang, W., **Gao, Riqiang ***, & Liao, Q. (2017). Weighted voting of discriminative regions for face recognition. *IEICE TRANSACTIONS on Information and Systems*, 100(11), 2734–2737.
- 33 **Gao, Riqiang**, Yang, W., Hu, X., & Liao, Q. (2016). Two-stage patch-based sparse multi-value descriptor for face recognition. *Visual Communications and Image Processing (VCIP)*, 1–4.
- 34 Yang, W., **Gao, Riqiang ***, Xu, Y., Sun, X., & Liao, Q. (2016). Discriminative patch-based sparse representation for face recognition. *IEEE International Conference on Signal Processing, Communications and Computing (ICSPCC)*.
- 35 **Gao, Riqiang**, Yang, W., Sun, X., Li, H., & Liao, Q. (2015). Locally collaborative representation in similar subspace for face recognition. *Chinese Conference on Biometric Recognition*, 88–95.

Officially Mentored Interns in Industry

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| 05/2024 - 09/2024 | <div style="display: flex; align-items: center;"> <div style="width: 15px; height: 15px; background-color: #800000; margin-right: 5px;"></div> <div>Thomas Li, MD.-Ph.D. candidate at Vanderbilt University, research intern at Siemens Healthineers</div> </div> |
| 06/2025 - now | <div style="display: flex; align-items: center; margin-bottom: 10px;"> <div style="width: 15px; height: 15px; background-color: #800000; margin-right: 5px;"></div> <div>Zihan Li, Ph.D. candidate at University of Washington, (full-time + part-time) research intern at Siemens Healthineers</div> </div> <div style="display: flex; align-items: center;"> <div style="width: 15px; height: 15px; background-color: #800000; margin-right: 5px;"></div> <div>Yuhan Wang, Ph.D. candidate at University of California, Santa Cruz, (full-time + part-time) research intern at Siemens Healthineers</div> </div> |

Officially Mentored Students at Vanderbilt

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|-------------------|---|
| 01/2020 - 05/2020 | <div style="display: flex; align-items: center;"> <div style="width: 15px; height: 15px; background-color: #800000; margin-right: 5px;"></div> <div>Qingyun Qian (master). First Job: Engineer in Huawei.</div> </div> |
| 01/2019 - 06/2019 | <div style="display: flex; align-items: center;"> <div style="width: 15px; height: 15px; background-color: #800000; margin-right: 5px;"></div> <div>Yiyuan Yang (bachelor). First Job: Engineer in Facebook.</div> </div> |
| 07/2019 - 09/2019 | <div style="display: flex; align-items: center;"> <div style="width: 15px; height: 15px; background-color: #800000; margin-right: 5px;"></div> <div>Lingfeng Li (bachelor). First Job: MS student in Northwestern University.</div> </div> |
| 01/2020 - 05/2020 | <div style="display: flex; align-items: center;"> <div style="width: 15px; height: 15px; background-color: #800000; margin-right: 5px;"></div> <div>Xinmeng Zhang (bachelor). First Job: Ph.D. student in Vanderbilt University.</div> </div> |

Some Academic Activities

Challenge Lead Organizer	■ GDP-HMM Grand Challenge at AAPM 2025
Invited Public Talk	■ MICCAI Industrial Talk Series (June 2025)
Program Committee	■ ICLR 2023 Workshop on Trustworthy Machine Learning for Healthcare ICCV2021 Workshop on Computer Vision for Automated Medical Diagnosis ICML 2021 Workshop Interpretable ML in Healthcare